

1905108

Exp 7-Multiplexers

Ans. to the Ques. No. 6

$$f = \Sigma (5, 6, 7, 11, 12, 13, 14, 15)$$

Truth table:

| A | B | C | D | f | Input in 2C74153 |
|---|---|---|---|---|------------------|
| 0 | 0 | 0 | 0 | 0 | $I_0 = 0$        |
| 0 | 0 | 0 | 1 | 0 |                  |
| 0 | 0 | 1 | 0 | 0 |                  |
| 0 | 0 | 1 | 1 | 0 |                  |
| 0 | 1 | 0 | 0 | 0 | $I_1 = C+D$      |
| 0 | 1 | 0 | 1 | 1 |                  |
| 0 | 1 | 1 | 0 | 1 |                  |
| 0 | 1 | 1 | 1 | 1 |                  |
| 1 | 0 | 0 | 0 | 0 | $I_2 = CD$       |
| 1 | 0 | 0 | 1 | 0 |                  |
| 1 | 0 | 1 | 0 | 0 |                  |
| 1 | 0 | 1 | 1 | 1 |                  |
| 1 | 1 | 0 | 0 | 1 | $I_3 = 1$        |
| 1 | 1 | 0 | 1 | 1 |                  |
| 1 | 1 | 1 | 0 | 1 |                  |
| 1 | 1 | 1 | 1 | 1 |                  |

Equation:

|   |   |   |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 0 | 0 |
| 1 | 0 | 0 |

$$I_0 = 0$$

|   |   |   |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 0 | 1 |
| 1 | 1 | 1 |

$$I_1 = C+D$$

|   |   |   |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 0 | 0 |
| 1 | 0 | 1 |

$$I_2 = CD$$

|   |   |   |
|---|---|---|
| 0 | 0 | 1 |
| 1 | 1 | 1 |
| 1 | 1 | 1 |

$$I_3 = 1$$

Input:

$$\text{Output: } f = m_5 + m_6 + m_7 + m_{11} + m_{12} + m_{13} + m_{14} + m_{15}$$

Input:  $I_0 = 0$ 

$$I_1 = C+D$$

$$I_2 = CD$$

$$I_3 = 1$$

Selector:

$$S_0 = B$$

$$S_1 = A$$